Dynamic Characterization of a Capacitive Deionization Cell for Integration with the Power Stage

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***Abstract*—This work proposes developing and presenting a novel methodology for the dynamic characterization of a Capacitive Deionization (CDI) cell —also applicable to CapMix systems—aimed at enabling its effective integration into a power conversion stage. The study is motivated by the increasing relevance of electrochemical desalination and energy harvesting technologies, and the need for accurate modelling tools to facilitate their inclusion in practical power electronic systems. The resulting Mathcad model is both simulation and experimentally validated and facilitates efficient coupling with power electronics for energy harvesting from salinity gradients, marking a significant advancement in CDI/CapMix system integration.**

Keywords—CapMix, Converter Topology, Dynamic characterization, Electrical Model.

I. Introduction

The global energy transition is currently at a critical juncture. According to the International Energy Agency (IEA) Global Energy Review 2025 [1], over 68% of the world's energy production still comes from fossil fuels like oil, natural gas, and coal. However, the steady growth of renewable energy sources is gradually reshaping this landscape. In 2024, renewables accounted for approximately 32% of the global energy mix, with a significant contribution from solar (25.2%), wind (5.5%), hydroelectric (1.1%) and others (0.2%) like biomass, geothermal energy as shown in Fig.1.

Among these sources, new alternatives with high potential are emerging [2], such as blue energy, which harnesses salinity gradients between solutions of differing salt concentrations—typically found where rivers meet the sea. The most studied technologies for converting this energy into electricity include three approaches, Pressure Retarded Osmosis (PRO), Reverse Electrodialysis (RED), and more recently, capacitive technologies called Capacitive Mixing (CapMix) cells, based on porous electrodes. The latter offers significant operational advantages, including simple design, low energy requirements, and modular scalability.

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CapMix cells generate electricity by exploiting the electrochemical potential difference that arises when alternating between high- and low-salinity solutions in contact with capacitive electrodes [3]. Although still in a developmental stage, this technology shows promising medium- and long-term potential due to its efficiency, low environmental impact, and adaptability to coastal environments and water treatment facilities.

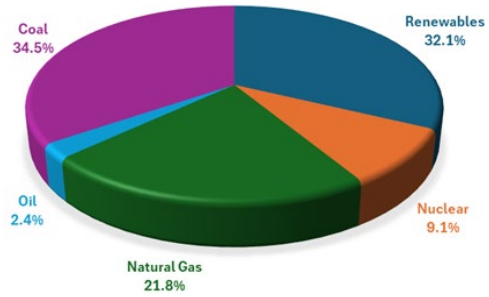


Fig.1. Electricity production in 2024.

Compared to other salinity gradient energy conversion technologies, such as PRO, RED, and salinity gradient fuel cells, CapMix offers certain advantages in terms of potential lower fouling sensitivity and simpler cell configurations. However, challenges persist, including optimizing energy extraction efficiency given losses inherent in cyclic capacitive processes and minimizing parasitic reactions. These factors emphasize the need for a detailed electrochemical characterization and energy extraction optimization in the development of CapMix cells [4].

CapMix technology evolves from foundational principles of capacitive deionization (CDI), which itself is an emerging technology traditionally aimed at water desalination through ion electrosorption. Early research illuminated that the charge storage mechanisms of CDI cells could be repurposed for energy harvesting via salinity gradients, leading to the concept of capacitive mixing cells. These CapMix cells combine the ion adsorption principles of CDI with cyclic salt/fresh water flows to enable electricity generation. The transition from desalination devices to energy harvesting units required a comprehensive understanding of the electrochemical and physical cell characteristics, leading to refined designs and control strategies [5].

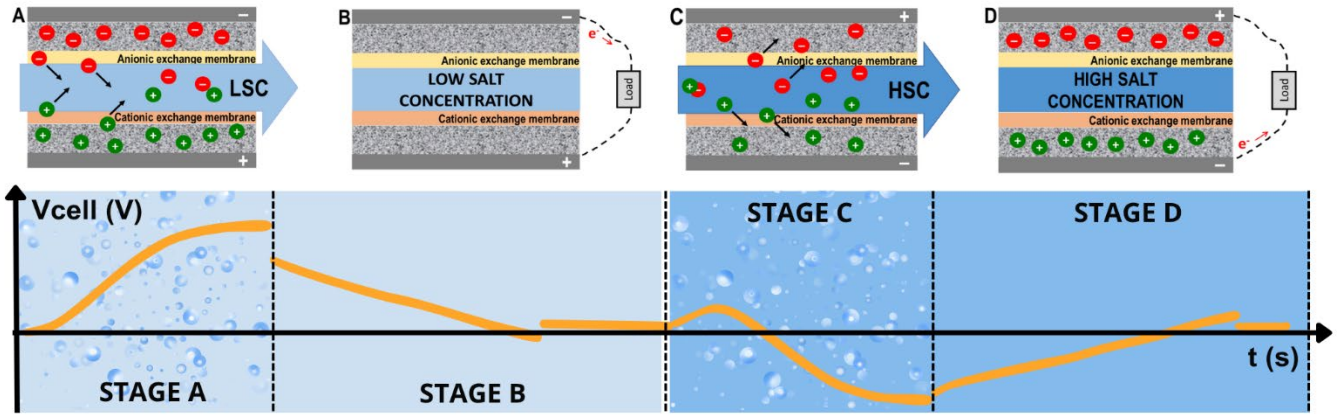


Fig. 2. The CapMix Cell operating principle.

Compared to traditional desalination technologies such as reverse osmosis and multi-stage flash distillation, CDI offers promising improvements in energy efficiency and lower operational costs, primarily due to its low pressure-driven flow requirements and the potential for energy recovery during electrode regeneration [6]. Performance metrics critical to evaluating CDI systems include salt adsorption capacity (SAC), which quantifies the mass of salt removed per unit mass of electrode material; charge efficiency, which denotes the fraction of the electrical charge effectively used to remove ions; and specific energy consumption (SEC), representing the amount of energy used per volume of treated water [7].

Over the past decades, industrial applications of CDI in water treatment laid substantial groundwork for scaling CapMix technologies. Developments in electrode materials, cell architectures, and system integration catalyzed CapMix research, linking it more explicitly with electrochemical energy storage systems such as supercapacitors. This conceptual integration situates CapMix as a complementary technology enhancing the capability of CDI cells beyond water treatment toward contributing renewable electricity from osmotic energy sources.

The operating principle of the CapMix cell is based on the Capacitive Donnan Potential (CDP), as shown in Fig. 2. When salt water flows through the cell, anions and cations are selectively adsorbed by electrodes via anion- and cation-exchange membranes, creating a voltage (Stage C). If connected to a load, electrons flow and discharge the cell (Stage D). When freshwater replaces the saltwater, the deposited ions diffuse back into the solution, reversing the potential (Stage A). Connecting a load now causes current to flow in the opposite direction (Stage A). By alternating salt and fresh water, continuous energy generation is achieved, as illustrated in Fig. 2. [8]

Given the relevance of this emerging technology, this paper presents three key aspects for evaluating its practical feasibility:

The design of a CapMix cell prototype, including its structural configuration, selected materials, number of electrodes, and functional architecture.

The electrical characterization of the system using a novel measurement technique, which allows detailed analysis of the cell's dynamic behavior and energy conversion capability.

The proposal of a specific power converter topology, designed to optimize the coupling of the CapMix cell with a DC bus or energy storage system, thereby improving overall system efficiency.

II. CapMix Cell

A. Concept and Design

CapMix technology stands out for its simplicity. It uses carbon-based porous electrodes immersed in electrolyte solutions of varying salinity. Energy is harvested by exploiting the change in electrode potential when alternating between saltwater and freshwater, without requiring redox reactions or moving parts. Currently, there are three main CapMix variants [9-10]:

- Capacitive Double Layer Expansion (CDLE): based on the variation of electrical double-layer capacitance when switching between high and low salinity solutions. It offers simple operation but relatively low power density.
- Flow-electrode CapMix (F-CapMix): employs circulating carbon slurry electrodes to enable continuous operation and enhanced scalability, at the cost of increased system complexity.
- Capacitive Donnan Potential (CDP): integrates ion-exchange membranes adjacent to the electrodes, leveraging Donnan potential differences induced by ion concentration gradients. This variant has demonstrated superior energy performance and higher power densities compared to the other two, and is currently considered the most promising CapMix configuration.

In parallel, Mixing Entropy Batteries (MEBs) have emerged as an alternative approach. These devices also exploit salinity gradients but through redox reactions in active electrode materials. While MEBs offer higher energy densities, they rely on specific redox materials and require more complex electrochemical control.

Due to its balance between performance, scalability, and simplicity, this work focuses on the development and evaluation of a CDP-type CapMix cell.

B. Electrodes

The electrodes employed in the CDP-type CapMix cell consist of a porous carbon material deposited onto graphite substrates, chosen for their low electrical resistance and chemical stability in aqueous environments. The carbon

material has been specifically developed and fabricated by INCAR-CSIC (Institute of Carbon Science and Technology, CSIC), with optimized properties for capacitive energy harvesting applications. (AX-7 Carbon) [11]

C. Configuration and Geometry

The CapMix prototype cell comprises a total of 10 electrodes arranged in a stacked configuration. Of these, the two electrodes at the outermost positions are coated with porous carbon (80x80x0.3mm) on a single face, while the remaining eight electrodes feature carbon layers on both sides. Each adjacent pair of electrodes forms an individual capacitor. Each electrode is in contact with a selective ion-exchange membrane, cation exchange membranes (CEM)(Fumasep FKS-30 10 x 10 from Fuel Cell Store) or anion exchange membranes (AEM) (Fumasep FAS-30 10 x 10 from Fuel Cell Store) which are not in direct contact with each other because a separator is used. In addition, this separator (0.45-mm thick (ED 64-102-086 from PC Cell) directs the flow of water. Finally, gaskets are used to prevent water leakage.

The dimensions of the electrodes have been selected to suit a laboratory-scale prototype (120x110x2mm), balancing ease of assembly with the ability to demonstrate scalability. The design is modular, enabling future expansion or optimization without fundamentally altering the operating principles of the cell. All electrodes are placed between two methacrylate blocks (160x160) machined to facilitate the entry and exit of water from the cell.

Fig. 3 illustrates the stacked configuration of the CapMix cell, designed to enhance system scalability and overall energy harvesting efficiency.

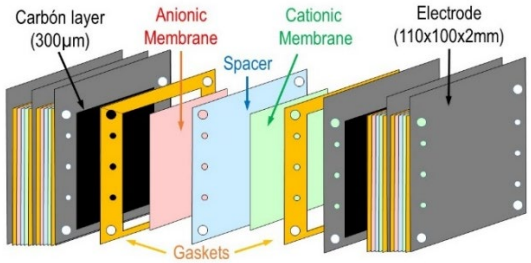


Fig.3. CapMix cell stacked configuration

III. Dynamic Characterization

The electrical characterization of CapMix cells has traditionally been performed under static or quasi-static conditions, wherein the cell is filled with water of a fixed salinity—either high or low—and subjected to a step current excitation. These tests, often conducted after ensuring homogeneity of the electrolyte, allow for the identification of basic electrical parameters (C, Rp and Rs) , and support simplified modeling of the cell as a double layer capacitor (DLC) [12]. While valuable as a first-order approximation, this methodology inherently neglects the time-dependent behavior arising from ion transport, adsorption hysteresis, and interfacial charge evolution.

However, in real-world CapMix applications, energy harvesting depends on the continuous alternation between high and low salt concentrated electrolytes, introducing a set of dynamic, and coupled phenomena not captured by static measurements. These include partial ion retention, non-equilibrium interfacial potentials, and diffusion-driven delays between fluid composition and electrode response. As a result, purely static characterizations can misrepresent both the energy available per cycle and the efficiency of real-time energy recovery.

Fig. 4 presents a comparative analysis of the voltage evolution in the high salt concentration (HSC) cell, highlighting the differences observed across various modelling approaches. When comparing the simulation results of this model ($Vcellsim_t$) with the actual experimental data ($Vreal_t$), a deviation was observed, which led to the proposal of a new electrical model for the cell ($Vcell_t$). This model includes an additional RC network, where C_{dyn} and R_{dyn} implement the exponential evolution shown in Fig. 4.

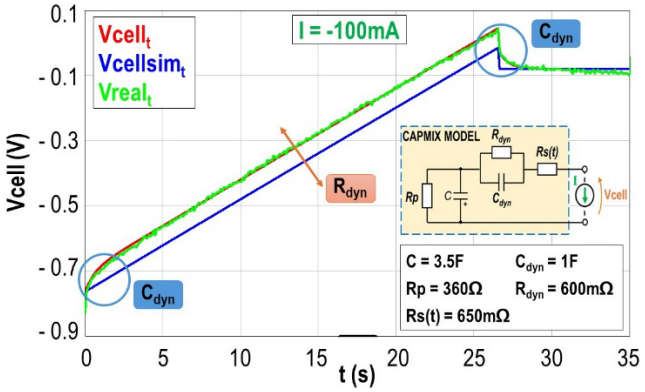


Fig.4. Comparative evolution of the HSC cell voltage, depending on the used model.

In this work, a dynamic characterization protocol is introduced to bridge this gap. The proposed method replicates the operational cycles of a CapMix system under realistic flow and switching conditions [4].

First, the cell is flushed with low salt concentration (LSC) water (1.5ms/cm), and the voltage is allowed to rise until it reaches a maximum open-circuit value.

A step current is then applied, discharging the cell and capturing the energy stored during the low-salinity phase.

Immediately after discharge, the electrolyte is switched to HSC water (55ms/cm), initiating a rapid change in ion distribution across the electrodes and membranes.

A second current pulse—of opposite polarity—is then applied, allowing the system to discharge again, now capturing energy from the high-salinity state.

As a result, the measurements inherently account for incomplete exchange, concentration gradients, residual charge, and internal redistribution of ions, offering a much closer approximation to the dynamic behavior of the cell in actual harvesting scenarios.

The dynamic tests performed reveal differentiated behaviors of the CapMix cell during low- and high-salinity cycles, highlighting that the cell's electrical properties are not constant, but instead vary according to the internal ionic state and the electrolyte conditions.

One of the most significant parameters in this regard is the series resistance, which has been found to differ between the low- and high-salinity phases. Specifically:

During the high-salinity cycle, the series resistance is noticeably lower due to the higher ionic conductivity of the electrolyte, which facilitates charge transport both in the solution and within the porous electrode structure. Moreover, under these conditions, the series resistance remains relatively

constant throughout the cycle, as the abundance of ions ensures stable conductivity and uniform charge movement.

In contrast, during the low-salinity cycle, the series resistance not only increases but also exhibits significant variability over time. This behavior is attributed to the lower ion concentration, which limits ionic conductivity. As the low-salinity cycle progresses, the series resistance tends to gradually decrease. This phenomenon is due to the release and redistribution of residual ions trapped within the porous carbon electrodes during the previous high-salinity cycle.

As summarized and illustrated in Fig. 5, three possibilities were considered: a maximum series resistance ($VcellRsmax_t$) an average resistance ($VcellRsave_t$) or a resistance depending on the voltage of the capacitor at any given moment ($Vcell_t$). These results were compared with the experimental results ($Vreal_t$), so the first was discarded due to the increase in error as the discharge progressed. The second was also discarded, as it was necessary to know the time of application of the current source to establish the average value, in addition to the greater error that was made.

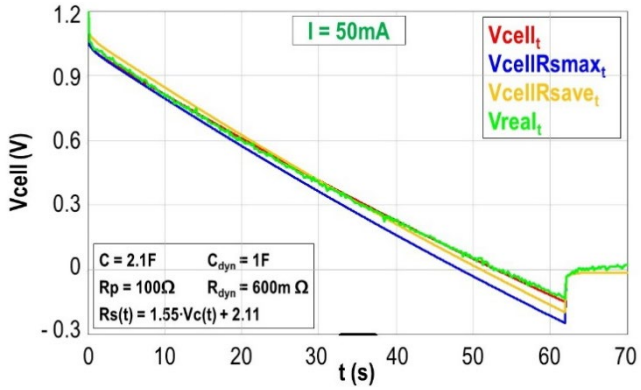


Fig.5. Comparative evolution of the LSC cell voltage, depending on used Rs value.

Another observation is the asymmetry in the voltage response between the low-salinity and high-salinity cycles. Even under identical current pulses (in magnitude but opposite direction), the voltage profiles differ in slope and amplitude and therefore the capacitance is different for each cycle.

To obtain the cell model, it is necessary to determine the output voltage. The circuit can be easily solved, it is possible to consider every capacitor plus parallel resistor driven by a current source as shown in Fig.6 (left). Thus, by applying Thevenin's theorem, it is possible to obtain a series RC circuit (right), whose solution is given by Eq. 1. This is valid when the power supply is connected ($t \leq toff$).

$$UcI(t) = U_0 - \left[I \cdot R \cdot \left(1 - e^{-\frac{t}{R \cdot C}}\right)\right] \quad (1)$$

When the current source is disconnected ($t > toff$), the voltage drop in each RC circuit is that imposed by Eq. 2 (discharge of a capacitor). While in Rs it is 0.

$$Uc0(t) = UcI(toff) \cdot \left(e^{-\frac{t-toff}{R \cdot C}}\right) \quad (2)$$

In this way, the model can be replaced by the one shown in Fig.6 at the bottom, wherein voltages Vc_{RpC} and Vc_{RCdyn} are known, and the voltage across the serial resistor is added.

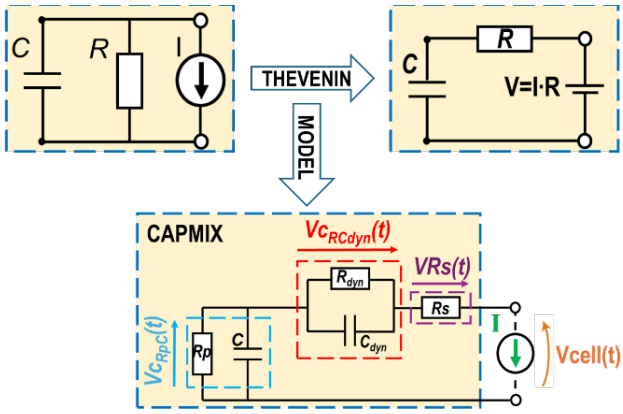


Fig.6. Equivalent circuit for CapMix modelling.

In this way, the cell output voltage for the new CapMix model can be calculated in Eq. 3.

$$Vcell(t) = Vc_{Rpc}(t) + Vc_{RC_{dyn}}(t) + VRs(t) \quad (3)$$

IV. Results

They are aimed at validating the proposed model in each case, so that a 'black box' is available when the converter is to be simulated. For this reason, a different current has been used in comparison to the one in the two previous graphs (Figs 4 and 5).

Belonging to the applied equations in section III, Tables 1 and 2 summarize the parameter values used in both the simulation ($Vcell_t$) and experimental stages ($Vreal_t$) to extract the output values. The obtained average value results (from Tables 1 and 2) for high (HSC) and low (LSC) cycles are shown in Figs 7 and 8, respectively.

TABLE 1. THE VALUES FOR CONSTANT PARAMETERS IN CAPMIX MODEL.

	Rp (Ω)	C (F)			Rdyn (Ω)	Cdyn (F)
		50 mA	100 mA	BOTH (Average)		
LSC	100	2.1	2.3	2.2	0.6	1
HSC	360	3.4	3.5	3.45		

TABLE 2. THE VALUES FOR SERIES RESISTANT IN CAPMIX MODEL.

	Rs(t)=A·Vc(t)+B	50mA	100mA	BOTH (average)
LSC	A	1.55	1.369	1.459
	B	2.11	1.826	1.968
HSC	Rs doesn't depend on Vc(t) so in all case Rs=0.65Ω			

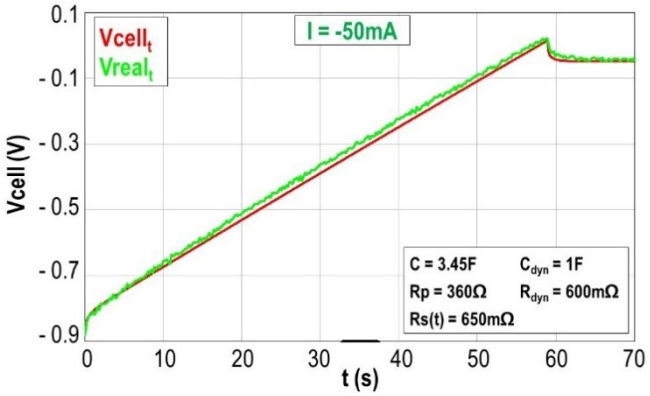


Fig.7. Comparative evolution of the HSC cell voltage.

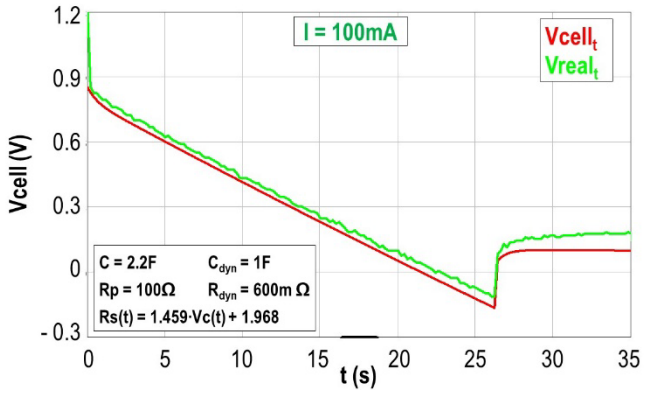


Fig.8. Comparative evolution of the LSC cell voltage.

V. Converter Topology

To effectively harvest the energy accumulated in the CapMix cell, it is necessary to integrate a DC-DC converter capable of adapting the variable voltage output of the cell to a stable and usable form. The proposed configuration in Fig. 9 includes a converter that allows two modes of operation: direct energy storage in batteries or injection into a DC bus operating at 48 V.

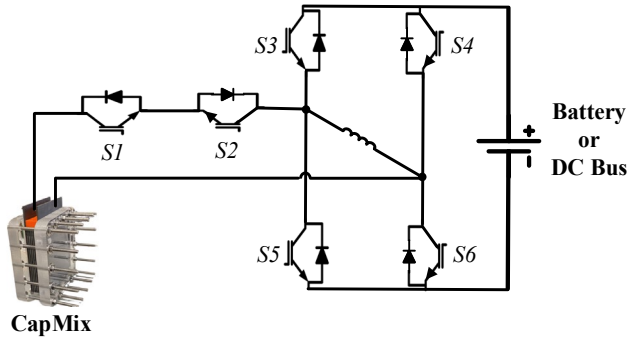


Fig.9. The proposed converter topology for CapMix.

The selection of a 48 V DC bus is not arbitrary. This voltage level offers an optimal balance between safety, efficiency, and compatibility. Moreover, 48 V has become an industry-standard in telecommunications, data centers, and emerging low-voltage microgrid applications [13-14], ensuring greater component availability and system scalability. Using a 48 V bus also facilitates direct compatibility with commercially available lithium-ion battery modules, which simplifies the storage architecture and reduces the need for additional conversion stages.

A. Requirements

Due to the low and alternating polarity of the voltage generated by the cell, a custom DC-DC step-up (boost) converter is required, tailored to the specific constraints of this application. Several critical design requirements have been identified:

- Step-Up Conversion (Boost Topology): The converter must elevate the cell's low output voltage to the 48 V level required by the DC bus. The typical output of the CapMix cell after each charge/discharge cycle is insufficient for direct use, making a high-efficiency step-up stage essential for energy recovery.
- Input Polarity Reversal Capability: Since the CapMix cell alternates its polarity with each cycle (positive in the low-salinity charging phase and negative in the high-salinity phase), the converter must be able to operate with both positive and negative input voltages.
- Inductor Decoupling During Discharge: To avoid additional energy losses during the discharge phase of the CapMix cell, the converter's inductor must be decoupled from the cell while the cell is releasing its stored energy. This measure prevents the series resistance of the inductor from degrading the overall energy extraction efficiency, which is particularly important due to the limited energy available per cycle.
- Unidirectional Power Flow:The converter does not need to support bidirectional power flow, simplifying the control and reducing hardware complexity. Its sole purpose is to extract energy from the cell and deliver it to the bus or battery.

In summary, the converter design must combine step-up capability, input polarity flexibility, and selective inductor coupling to ensure maximum energy transfer efficiency under the unique dynamic and electrical characteristics of the CapMix system.

B. Operation Modes

In stages B and D in Fig. 2 the energy is extracted, and therefore, the converter is necessary. In Fig. 10, as we have the positively charged cell (LSC) the coil will be charged, and then the energy will be transferred to the battery (Fig. 10).

Fig. 11 represents the same, but taking into account that the cell is negatively charged (HSC).

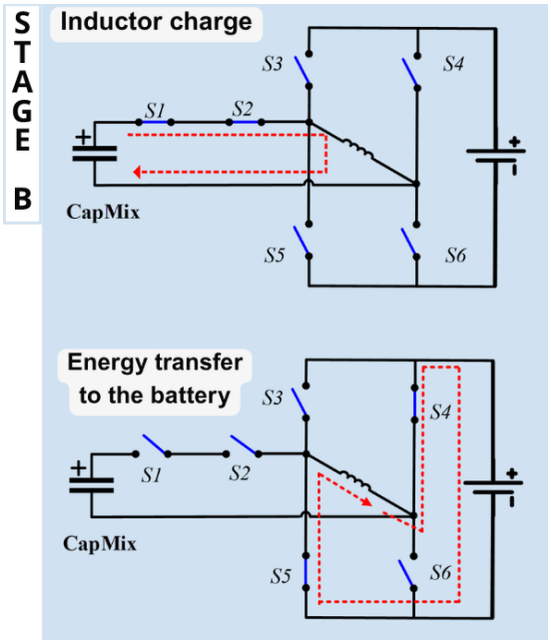


Fig.10. Converter operation in stage B.

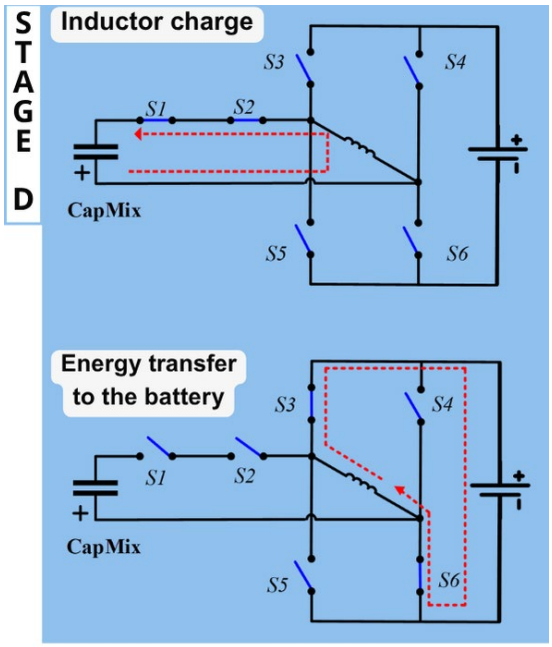


Fig.11. Converter operation in stage D.

VI. Conclusion

A novel methodology for the dynamic characterization of a CDI cell applicable to CapMix systems for the power conversion stage has been investigated in this paper. The proposed approach enables more effective integration of these technologies into power conversion stages, representing a significant advancement in the energy harvesting from salinity gradients.

A more comprehensive mathematical model has been developed, improving upon previous approximations, and its results have been validated through both simulation and experimental data. The characterization has proven robust across different discharge current conditions, which reinforces the reliability of the proposed model.

Furthermore, standardized electrical parameters have been established, allowing the cell to be treated as a “black box” from the converter’s perspective. This simplifies integration without compromising accuracy. Based on this, a novel converter topology has been designed, specifically optimized for the dynamic behavior of the cell, and its operating mode has been described in detail. These results lay the groundwork for the development of more efficient and modular energy conversion systems in energy harvesting applications, facilitating the future practical implementation of CDI/CapMix technologies in real-world environments.

References

[1] International Energy Agency. (2025). Global energy review 2025. https://www.iea.org/reports/global-energy-review-2025

[2] Cordis, C. (2013, 21 febrero). Salinity gradient energy provides green solution. CORDIS | European Commission. https://cordis.europa.eu/article/id/90491-salinity-gradient-energy-provides-green-solution.

[3] Lobato, B., Flores, S. L., dos Santos-Gómez, L., García, A. B., Pernía, A. M., Prieto, M. J., Busto, M. G., & Arenillas, A. (2024). Designing and Optimizing Electrode Materials for Energy Harvesting in CAPMIX Cells. Nanomaterials, 14(24), 2031. https://doi.org/10.3390/nano14242031

[4] M. G. Busto, M. J. Prieto, J. A. Martín-Ramos, J. A. Martínez and A. M. Pernía, "Optimization of the Energy Extraction Process From Capacitive Deionization Cells," in IEEE Transactions on Industrial Electronics, vol. 72, no. 7, pp. 7084-7093, July 2025, doi: 10.1109/TIE.2024.3522505

[5] A. M. Pernía, M. J. Prieto, J. A. Martín-Ramos, P. J. Villegas, and F. J. Álvarez-González, ‘Energy Recovery in Capacitive Deionization Technology’, Desalination and Water Treatment, 37-54. Sept. 19, 2018. doi: 10.5772/intechopen.75537.

[6] Abumadi, F. A., Koujan, M., Laoui, T., Atieh, M. A., & Khalil, K. A. (2023). Capacitive Deionization Water Desalination Technology, Process Optimization and Cost Analysis - A Review. Advances In Science And Technology, 129, 19-25. https://doi.org/10.4028/p-wo4cjg

[7] Liu, M., He, M., Han, J., Sun, Y., Jiang, H., Li, Z., Li, Y., & Zhang, H. (2022). Recent Advances in Capacitive Deionization: Research Progress and Application Prospects. Sustainability, 14(21), 14429. https://doi.org/10.3390/su142114429

[8] M. G. Busto, M. J. Prieto, A. M. Pernía, A. Arenillas and J. R. Álvarez, "Optimization of Energy Harvesting in CapMix Cells," 2022 IEEE 1st Industrial Electronics Society Annual On-Line Conference (ONCON), kharagpur, India, 2022, pp. 1-6, doi: 10.1109/ONCON56984.2022.10126727.

[9] Z. Zou, L. Liu, S. Meng, and X. Bian, “Comparative study on the performance of capacitive mixing under different operational modes,” Energy Reports, vol. 8, pp. 7325–7335, 2022. https://doi.org/10.1016/j.egyr.2022.05.245

[10] G. R. Iglesias, S. Ahualli, M. M. Fernández, M. L. Jiménez,and Ángel V. Delgado, “Stacking of capacitive cells for electrical energy production by salinity exchange,” Journal of Power Sources, vol. 318, pp. 283–290, 2016. https://doi.org/10.1016/j.jpowsour.2016.04.026.

[11] B. Lobato, S. L. Flores, L. dos Santos-Gómez, A. B. García, A. M. Pernía, M. J. Prieto, M. G. Busto y A. Arenillas, “Designing and Optimizing Electrode Materials for Energy Harvesting in CAPMIX Cells,” Nanomaterials, vol. 14, no. 24, art. no. 2031, 18-Dic-2024. doi: 10.3390/nano14242031

[12] J. Wu, “Understanding the electric double-layer structure, capacitance, and charging dynamics,” Chemical Reviews, vol. 122, no. 12, pp. 10 821–10 859, 2022, pMID: 35594506. [Online]. Available: https://doi.org/10.1021/acs.chemrev.2c00097

[13] Ou, K., Yuan, W.-W., & Kim, Y.-B. (2021). Development of optimal energy management for a residential fuel cell hybrid power system with heat recovery. Energy (Oxford, England), 219(119499), 119499. https://doi.org/10.1016/j.energy.2020.119499

[14] Andrade, P., Alcaso, A. N., Bento, F., & Marques Cardoso, A. J. (2022). Buck-Boost DC-DC Converters for Fuel Cell Applications in DC Microgrids—State-of-the-Art. Electronics, 11(23), 3941. https://doi.org/10.3390/electronics11233941.